

# Implementation-Risk Differentials in Hydrogen Technology Pathways:

*Evidence from a Segmented Investment Ecosystem and Carbon-Price-Conditional Risk*

Sake Saakstra

Vrije Universiteit Amsterdam — Department of Econometrics

Gasunie — Business Line Waterstof Nederland

s.saakstra@vu.nl

Working Paper — May 2026

## Abstract

Hydrogen project realization is not primarily a comparison of two interchangeable technologies but a process governed by **structurally segmented investment ecosystems** within which implementation risk concentrates on **terminal cancellation rather than reversible delay** and is conditional on carbon-price exposure. Using the IEA Hydrogen Projects Database covering 2,380 announced projects globally, we develop this view through two primary contributions, supported by convergent empirical evidence and a mechanistic interpretation.

*Primary contribution 1 — Segmented investment ecosystems.* Blue\_CCS and PEM projects operate in measurably distinct investment ecosystems: the propensity model achieves McFadden  $R^2 = 0.70$ ; propensity score matching uses only 29 unique PEM controls for 147 Blue\_CCS matches; entropy balancing collapses to an effective sample size of 8. We argue that this non-exchangeability is substantive content rather than a methodological obstacle to reweight away.

*Primary contribution 2 — Terminal cancellation, not reversible delay.* The Fine-Gray competing-risks decomposition reveals that the Blue\_CCS implementation-risk differential concentrates almost entirely on terminal cancellation ( $HR = 13.19 [5.28, 32.91]$ ) rather than on real-option delay ( $HR = 1.20 [0.34, 4.26]$ , indistinguishable from one). Within the observed sample, Blue\_CCS projects appear to terminate rather than enter a pause-and-wait state.

*Supporting evidence — convergence across estimators.* Within the comparable subset, eleven estimators across four classes of identifying assumptions yield Blue\_CCS hazard ratios consistently between 5 and 14 (preferred doubly robust  $HR = 6.87$ ; Cox PH =  $11.93 [4.67, 30.49]$ ); proportional hazards is supported (Schoenfeld  $p = 0.59$ ), sponsor frailty is absent ( $\theta = 0.000$ ), and leave-one-region-out analyses confirm regional robustness across all seven regional exclusions ( $p < 0.0001$  throughout). The contribution lies in the convergence of these independent signals toward a common qualitative pattern, not in precise effect-size identification, which 43 events would not support.

*Mechanism — carbon-price-conditional risk.* The Blue\_CCS  $\times$  EUA interaction coefficient is  $-2.51$  ( $p < 0.0001$ ), implying steeply declining predicted hazard ratios as carbon prices

rise, with gas-price, VIX, and policy-uncertainty interactions all statistically insignificant. Predictions at low-carbon-price extrapolations should be read as directional stress-test indications rather than literal point forecasts.

*Conceptual interpretation.* The findings are consistent with fiscal support for blue hydrogen (the IRA 45V tax credit, EU REPowerEU support, the RFNBO Delegated Acts) being interpretable as transition-period insurance against low-carbon-price scenarios in which the supported technology would not survive to benefit from later recovery.

**Keywords:** hydrogen, carbon capture, project cancellation, transition risk, segmented investment ecosystems, terminal cancellation, real-option delay, carbon-price-conditional risk, Fine-Gray competing risks, propensity score, doubly robust estimation, entropy balancing

**JEL classification:** Q42, Q48, C41, L94, G31

# 1. Introduction

Decarbonisation of energy systems requires building infrastructure that did not exist at scale a decade ago. Hydrogen produced via electrolysis is widely seen as a critical vector for hard-to-abate sectors including steel, ammonia, refining, and long-distance transport (IEA, 2023; Glenk and Reichelstein, 2019). The EU REPowerEU package (European Commission, 2022) commits to 20 million tonnes of renewable hydrogen by 2030, the US Inflation Reduction Act (US Congress, 2022) provides up to \$3 per kilogram production tax credit via the 45V provision, and the EU Renewable Fuels of Non-Biological Origin (RFNBO) Delegated Acts (European Commission, 2023) codify the eligibility criteria for clean hydrogen. Yet a growing literature documents that announced infrastructure projects translate into operational facilities at low rates (Flyvbjerg, 2014), and the transition-finance literature has begun to quantify the implications of project failure for portfolio-level transition risk (Mercure et al., 2018; Bolton and Kacperczyk, 2021; Caldecott, 2018).

Within the hydrogen sector, a fundamental question concerns the relative reliability of two distinct technology pathways. Blue hydrogen — produced from natural gas with carbon capture and storage (CCS) — leverages existing fossil infrastructure and can in principle be deployed at scale rapidly (Bui et al., 2018; Budinis et al., 2018), but depends on functioning CCS capacity, faces carbon-price exposure, and confronts reputational risks. Green hydrogen via proton-exchange-membrane (PEM) electrolysis is purely electrical and pairs with renewable generation (Glenk and Reichelstein, 2019), but is currently more capital-intensive and requires substantial electrical infrastructure.

## 1.1. Central thesis and contribution hierarchy

This paper argues a single thesis: *hydrogen project realisation is not primarily a comparison of two interchangeable technologies but a process governed by structurally segmented investment ecosystems within which implementation risk concentrates on terminal cancellation rather than reversible delay, and is conditional on carbon-price exposure*. We develop this view through two primary contributions, supported by convergent empirical evidence and a mechanistic interpretation.

**Primary contribution 1 — Segmentation.** Blue\_CCS and PEM projects operate in measurably distinct investment ecosystems. The propensity score model that predicts technology choice from observable project characteristics achieves McFadden pseudo- $R^2 = 0.698$ . Propensity score matching with 0.05 caliper uses only 29 unique PEM controls to match 147 Blue\_CCS projects, and entropy balancing concentrates effective weight on just 8 of 470 PEM observations before diverging into non-identification. We treat this non-exchangeability as substantive content rather than a methodological problem to reweight away.

**Primary contribution 2 — Terminal cancellation, not reversible delay.** The Fine-Gray competing-risks decomposition reveals that the Blue\_CCS implementation-risk differential is concentrated almost entirely on terminal cancellation (HR = 13.19, 95% CI [5.28, 32.91],  $p < 0.0001$ ) rather than on real-option delay (HR = 1.20, 95% CI [0.34, 4.26],  $p = 0.78$ ). Within the observed sample, Blue\_CCS projects appear to terminate rather than enter a pause-and-wait

state in response to deteriorating conditions.

**Supporting evidence — Convergent estimates across methods.** Within the comparable subset, eleven estimators across four classes of identifying assumptions yield Blue\_CCS hazard ratios consistently between 5 and 14, with leave-one-region-out analyses confirming the qualitative finding is regionally robust (HR range [11.1, 19.9] across seven LOR specifications,  $p < 0.0001$  throughout). The preferred doubly robust marginal estimate is  $\text{HR} = 6.87$ ; the Cox PH estimate is  $\text{HR} = 11.93$  [4.67, 30.49]. Proportional hazards is supported (Schoenfeld  $p = 0.59$ ); sponsor frailty is absent ( $\theta = 0.000$ ). The contribution here lies in *convergence*: independent estimators with different identifying assumptions, different sampling structures, and different functional forms point in the same qualitative direction — a stronger claim, given 43 events, than any single estimator’s point estimate could support.

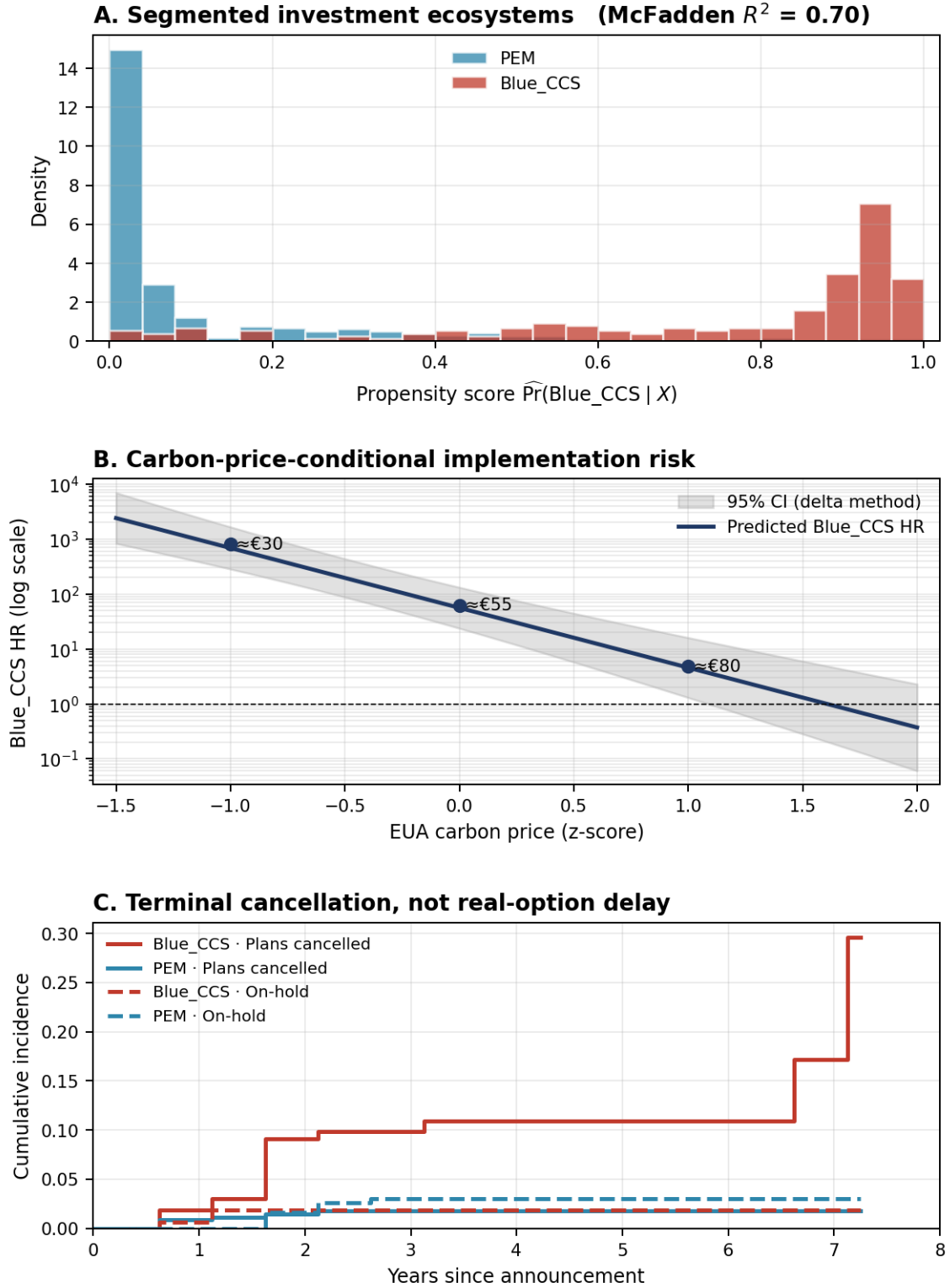
**Mechanism — Carbon-price-conditional risk.** The  $\text{Blue\_CCS} \times \text{EUA}$  carbon-price interaction coefficient is  $-2.51$  ( $p < 0.0001$ ). Delta-method marginal effects show that the predicted Blue\_CCS hazard ratio declines steeply as carbon prices rise. Interactions with gas prices, VIX, and economic policy uncertainty are statistically insignificant, sharpening the interpretation: it is not generalised macro stress that mechanistically drives the Blue\_CCS relative cancellation hazard, but specifically carbon-price exposure. Extreme predictions at very low carbon-price extrapolations are best read as directional stress-test indications rather than literal point forecasts.

## 1.2. Contribution and scope

The two primary contributions distinguish this paper from existing treatments of hydrogen technology comparison in two ways. First, methodologically: where prior work has compared blue versus green hydrogen failure rates with the implicit assumption that the two technologies represent comparable subpopulations, we document that the two populations are not exchangeable in any operational sense — and that this non-exchangeability is itself substantive content. Second, substantively: the terminal-versus-delay decomposition combined with the carbon-price-conditional mechanism is interpretable, conceptually, as suggesting that fiscal support schemes for blue hydrogen (the IRA 45V tax credit, EU REPowerEU support, the RFNBO Delegated Acts) function less as cost-of-capital subsidies and more as transition-period insurance against low-carbon-price scenarios in which the supported technology would not survive to benefit from later carbon-price recovery. We position this as a conceptual interpretation of our findings, not as a normative policy conclusion — the data structure does not support a full welfare or policy-optimisation analysis.

## 1.3. Roadmap

Section 2 describes the data. Section 3 documents the segmented-ecosystem structure (pillar 1). Section 4 outlines the empirical strategy across eleven estimators. Section 5 presents results across pillars 2–4. Section 6 reports additional robustness checks including leave-one-region-out analyses. Section 7 discusses identification, internal validity, and substantive implications. Section 8 concludes.



**Figure 1: Three-panel headline summary.** *Panel A* — propensity score distributions for Blue\_CCS and PEM; the bimodal distributions and high McFadden  $R^2$  visualise pillar (1), segmentation. *Panel B* — predicted Blue\_CCS hazard ratio as a function of EUA carbon price (z-score, log scale on y-axis); the steep decline of the curve and the 95% delta-method CI band visualise pillar (4), carbon-price-conditional. *Panel C* — Aalen-Johansen cumulative incidence by treatment group and event type; the divergence between the Blue\_CCS terminal-cancellation curve and all other curves visualises pillar (3), terminal-not-pause. Pillar (2), eleven-estimator robustness, is documented in Table 1.

## 2. Data

The IEA Hydrogen Projects Database (IEA, 2024) (March 2024 release) lists 2,380 projects globally with fields for technology, capacity, primary owner, geography, year of announcement, expected date online, and project status. We distinguish two failure events: *Plans cancelled* (terminal failure, irreversible) and *On-hold (confirmed)* (real-option delay, potentially reversible). The competing-risks specification treats these as separate causes (Fine and Gray, 1999).

We restrict the sample to projects with valid announcement years and capacity data. After filtering, the working sample comprises 714 projects (244 Blue\_CCS, 470 PEM) with 43 events (31 cancellations, 12 on-hold). Person-year expansion (Allison, 2014; Singer and Willett, 2003) yields 4,246 observations. Time-varying macro covariates come from European Energy Exchange (EUA), TTF gas futures, the VIX, the Baker-Bloom-Davis EPU index (Baker et al., 2016), ENTSO-E day-ahead power prices, and the World Bank WDI. Data sources are documented in Appendix B; all replication code is available at the repository linked in the Data Availability statement.

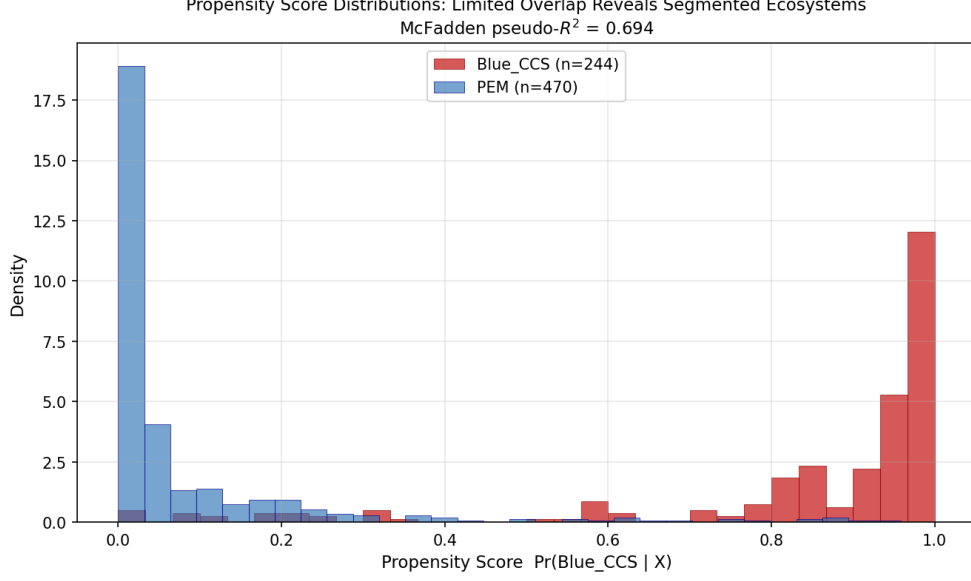
## 3. Segmented Investment Ecosystems: Descriptive Evidence

### 3.1. Differential project characteristics

Blue\_CCS projects are an order of magnitude larger on average (mean capacity 1,850 MW vs 290 MW), more concentrated in oil-major and industrial-gas sponsors, and more frequently located in North America and the Middle East and North Africa. PEM projects skew toward smaller utility and pure-play sponsors and concentrate in Europe and Asia.

### 3.2. Propensity score diagnostics

A logistic regression of the Blue\_CCS indicator on log capacity, year of announcement, region, and sponsor type yields McFadden pseudo- $R^2 = 0.698$  (McFadden, 1974). Values above 0.5 are uncommon in the causal inference literature (Imbens and Rubin, 2015) and typically indicate selection so strong that the treatment and control populations are not exchangeable in a substantive sense (Rosenbaum and Rubin, 1983). The propensity score range for Blue\_CCS is [0.000, 1.000] with median 0.947; for PEM it is [0.000, 0.962] with median 0.017.



**Figure 2: Propensity score distributions by treatment status.** Bimodal distributions with limited overlap support the segmented-ecosystem framing.

### 3.3. Convergent evidence from two reweighting methods

Two methodologically distinct reweighting approaches independently confirm the limited overlap. Propensity score matching (Rosenbaum and Rubin, 1983; Caliendo and Kopeinig, 2008) with 0.05 caliper produces 147 matched pairs but relies on only 29 unique PEM control observations, each used on average five times. Entropy balancing (Hainmueller, 2012) fails to find an interior solution: it concentrates effective weight on approximately 8 of 470 PEM controls (ESS = 1.7%) and the corresponding weighted regression coefficient diverges to non-identification. This dual failure provides strong evidence that no reweighting strategy can transform the two technology populations into exchangeable groups.

### 3.4. Interpretation

The combined evidence supports a substantive conclusion: Blue\_CCS and PEM hydrogen projects are not random draws from a single project population to which one of two technology labels has been assigned. They are structurally different project types embedded in different investment ecosystems.

## 4. Empirical Strategy

We employ eleven estimators across four classes of identifying assumptions.

### 4.1. Discrete-time hazard model

The baseline discrete-time hazard model in person-period form (Allison, 2014; Singer and Willett, 2003) is

$$h_{it} = \Pr(T_i = t \mid T_i \geq t, X_{it}) = \Lambda(\alpha_t + \beta \cdot \text{Blue\_CCS}_i + X'_{it}\gamma), \quad (1)$$

where  $\Lambda(\cdot)$  is the logistic link,  $\alpha_t$  parametrises the baseline hazard via a polynomial in years-since-start,  $X_{it}$  contains the baseline covariates (log capacity, region, sponsor type) and time-varying macro covariates, and  $\beta$  is the coefficient of interest. We report cluster-robust standard errors at the sponsor and sponsor  $\times$  region levels following [Wooldridge \(2010\)](#).

## 4.2. Cox proportional hazards

The Cox proportional hazards model ([Cox, 1972](#); [Andersen and Gill, 1982](#)) serves as a continuous-time robustness check:

$$\lambda_i(t | X_i) = \lambda_0(t) \exp(X_i' \beta). \quad (2)$$

We test the proportional-hazards assumption using Schoenfeld residuals ([Schoenfeld, 1982](#); [Grambsch and Therneau, 1994](#)) with rank time transformation.

## 4.3. Propensity-based selection corrections

Three propensity-based corrections address selection on observables. *Propensity score matching* uses one-to-one nearest-neighbour matching with caliper 0.05 ([Caliendo and Kopeinig, 2008](#)). *Inverse probability weighting* ([Robins et al., 1994](#)) uses stabilised weights with PS trimming. The *doubly robust* (augmented IPW) estimator combines outcome regression with propensity weighting:

$$\hat{\tau}_{\text{DR}} = \frac{1}{N} \sum_{i=1}^N \left[ \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right], \quad (3)$$

which is consistent if either the propensity model  $\hat{e}(X)$  or the outcome regressions  $\hat{\mu}_T(X)$  are correctly specified ([Bang and Robins, 2005](#)).

## 4.4. Entropy balancing

Entropy balancing ([Hainmueller, 2012](#)) solves for control weights that satisfy exact moment constraints on observable covariates while remaining as close as possible to a uniform prior in Kullback-Leibler distance:

$$\min_{\{w_i\}_{i:T_i=0}} \sum_{i:T_i=0} w_i \log(w_i/q_i) \quad \text{s.t.} \quad \sum_{i:T_i=0} w_i c_k(X_i) = m_k \quad \forall k, \quad \sum_{i:T_i=0} w_i = 1. \quad (4)$$

## 4.5. Firth's penalised likelihood

[Firth \(1993\)](#) introduces a penalty based on the Jeffreys prior to address finite-sample bias and complete separation ([Heinze and Schemper, 2002](#)):

$$\ell_{\text{Firth}}(\beta) = \ell(\beta) + \frac{1}{2} \log |\mathcal{I}(\beta)|, \quad (5)$$

where  $\mathcal{I}(\beta)$  is the Fisher information matrix. The penalty yields finite estimates even under complete separation and reduces small-sample bias of order  $O(1/n)$ .



#### 4.6. Fine-Gray subdistribution hazards

The Fine-Gray model (Fine and Gray, 1999) estimates the subdistribution hazard for each competing risk  $k$ :

$$\lambda_k^{\text{sd}}(t) = -\frac{d \log[1 - F_k(t)]}{dt}, \quad F_k(t) = \Pr(T \leq t, J = k), \quad (6)$$

where  $J$  is the cause indicator. The subdistribution hazard for cause  $k$  is interpreted relative to a risk set that retains subjects already experiencing competing events, enabling direct estimation of the cumulative incidence function.

#### 4.7. Shared frailty Cox

To address sponsor-level heterogeneity beyond the fixed sponsor type indicator, we estimate a shared frailty Cox model with gamma-distributed random effects:

$$\lambda_{ij}(t | X_{ij}, u_j) = \lambda_0(t) \exp(X'_{ij}\beta) u_j, \quad u_j \stackrel{\text{iid}}{\sim} \Gamma(1/\theta, 1/\theta). \quad (7)$$

The frailty variance  $\theta$  quantifies unobserved within-sponsor-type heterogeneity (Therneau et al., 2003).

#### 4.8. Macro-financial interactions and marginal effects

To position the analysis in transition-finance territory, we estimate four interaction models of the form

$$h_{it} = \Lambda(\alpha_t + \beta_1 \text{Blue\_CCS}_i + \beta_2 Z_{it} + \beta_3 (\text{Blue\_CCS}_i \times Z_{it}) + X'_{it}\gamma), \quad (8)$$

where  $Z_{it}$  is one of: standardised EUA carbon price, TTF gas price, VIX, or EPU index. The predicted Blue\_CCS log-hazard ratio as a function of  $z$  is the linear combination

$$\widehat{\text{LHR}}(z) = \hat{\beta}_1 + \hat{\beta}_3 z, \quad (9)$$

whose variance follows from the delta method (Wooldridge, 2010):

$$\widehat{\text{Var}}(\widehat{\text{LHR}}(z)) = \widehat{\text{Var}}(\hat{\beta}_1) + z^2 \widehat{\text{Var}}(\hat{\beta}_3) + 2z \widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_3). \quad (10)$$

The 95% confidence interval for the predicted hazard ratio at level  $z$  is  $\exp\left(\widehat{\text{LHR}}(z) \pm 1.96 \sqrt{\widehat{\text{Var}}(\widehat{\text{LHR}}(z))}\right)$ .

### 5. Results

#### 5.1. Headline estimates across methods

Table 1 reports the Blue\_CCS hazard ratio under all eleven estimators. The estimates that yield interior solutions cluster between approximately 5 and 14, with the doubly robust marginal estimate at HR = 6.87 as our preferred specification.

**Table 1:** Blue\_CCS hazard ratio across eleven estimators.

| Estimator                               | Blue_CCS HR   | 95% CI / SE   | $p$        | Sample      |
|-----------------------------------------|---------------|---------------|------------|-------------|
| GLM, iid SE                             | 14.32         | SE 0.506      | $< 0.0001$ | 4,246 p-yr  |
| GLM, cluster sponsor SE                 | 14.32         | SE 0.328      | $< 0.0001$ | 4,246 p-yr  |
| GLM, cluster sponsor $\times$ region SE | 14.32         | SE 0.490      | $< 0.0001$ | 4,246 p-yr  |
| Cox PH                                  | 11.93         | [4.67, 30.49] | $< 0.0001$ | 714 proj.   |
| PSM, caliper 0.05                       | 5.22          | SE 1.09       | 0.130      | 992 p-yr    |
| IPW (stabilised)                        | 14.50         | SE 0.51       | $< 0.0001$ | 3,706 p-yr  |
| <b>Doubly Robust (preferred)</b>        | <b>6.87</b>   | —             | $< 0.0001$ | 3,706 p-yr  |
| Entropy Balancing                       | Failed (sep.) | ESS = 8/470   | —          | 714 proj.   |
| Firth penalised baseline                | 11.74         | SE 0.38       | $< 0.0001$ | 12,354 p-yr |
| <b>Fine-Gray, Plans cancelled</b>       | <b>13.19</b>  | [5.28, 32.91] | $< 0.0001$ | 714 proj.   |
| <b>Fine-Gray, On-hold</b>               | <b>1.20</b>   | [0.34, 4.26]  | 0.782      | 714 proj.   |
| Shared Frailty Cox ( $\gamma$ )         | 13.73         | [5.24, 35.98] | $< 0.0001$ | 714 proj.   |

Notes: Cox concordance index = 0.810. Shared frailty variance  $\theta = 0.000$  indicates that the fixed sponsor type indicator captures all systematic sponsor-level variation.

## 5.2. Cox proportional hazards and Schoenfeld test

The Cox proportional hazards model yields Blue\_CCS HR = 11.93 (95% CI [4.67, 30.49]) with concordance index 0.810. The Schoenfeld residual test with rank time transformation supports the proportional hazards assumption for both Blue\_CCS ( $p = 0.585$ ) and log capacity ( $p = 0.281$ ). This justifies reporting a single hazard ratio that summarises the technology differential across the project lifecycle without requiring time-stratified specifications.

## 5.3. Cluster-robust standard errors

Naive (iid) SE for the Blue\_CCS coefficient is 0.506. Cluster-robust SE at the sponsor level is 0.328, and at the sponsor  $\times$  region level is 0.490. The sponsor-level cluster SE is *smaller* than the iid SE rather than larger, reflecting that many sponsors in our sample have only one project (singleton clusters), particularly among pure-play hydrogen firms. The sponsor  $\times$  region clustering yields a cluster SE closer to the iid estimate, consistent with this interpretation. The Blue\_CCS coefficient remains strongly significant under all clustering schemes ( $p < 0.0001$  throughout).

## 5.4. Marginal effects: carbon-price-conditional risk

The macro-financial interaction analysis is the central substantive contribution. Table 2 reports the four interaction models; only the EUA interaction is statistically significant.

**Table 2:** Macro-financial interaction effects on Blue\_CCS hazard.

| Interaction term                                    | Blue main       | Macro main       | Interaction      | Int. $p$           |
|-----------------------------------------------------|-----------------|------------------|------------------|--------------------|
| Blue_CCS $\times$ TTF_gas <sub><math>z</math></sub> | 2.582***        | −0.122***        | −0.670           | 0.315              |
| Blue_CCS $\times$ EUA <sub><math>z</math></sub>     | <b>4.026***</b> | <b>+2.516***</b> | <b>−2.507***</b> | <b>&lt; 0.0001</b> |
| Blue_CCS $\times$ VIX <sub><math>z</math></sub>     | —               | —                | +0.246           | 0.569              |
| Blue_CCS $\times$ EPU <sub><math>z</math></sub>     | —               | —                | +0.006           | 0.986              |

Notes: Each row is a separate hazard regression on the full person-year panel ( $n = 4,246$ ) with cluster-robust SE at sponsor level. Macro variables are standardised. Significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Figure 3 visualises the predicted Blue\_CCS cancellation hazard ratio as a function of EUA carbon price (z-score), with 95% confidence intervals computed via the delta method (Equation 10).

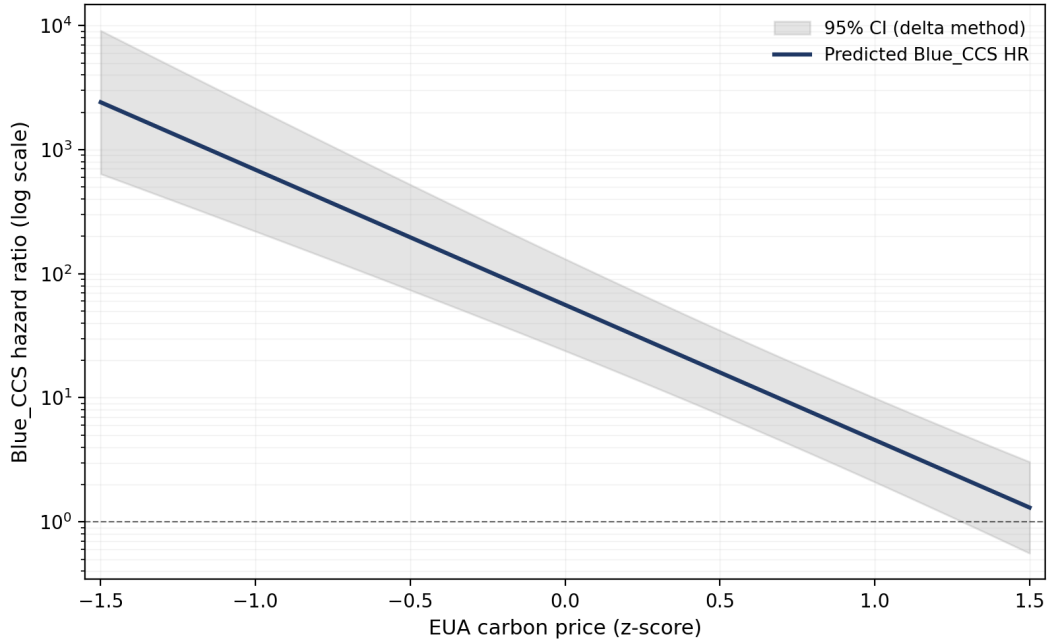
**Figure 3:** Predicted Blue\_CCS cancellation hazard ratio as a function of EUA carbon price. Shaded band: 95% CI from delta method. Approximate EUA price levels corresponding to z-scores are given in Table 3.

Table 3 reports the predicted Blue\_CCS hazard ratios at key carbon-price levels with full 95% confidence intervals.

**Table 3:** Predicted Blue\_CCS hazard ratio at key carbon-price levels.

| EUA z-score                     | Approx. EUA (€/tCO <sub>2</sub> ) | Predicted Blue_CCS HR | 95% CI               |
|---------------------------------|-----------------------------------|-----------------------|----------------------|
| $z = -1$ (low carbon)           | $\approx 30$                      | 673.26                | [215.42, 2,104.17]   |
| $z = 0$ (historical mean)       | $\approx 55$                      | 59.73                 | [25.33, 140.85]      |
| $z = +1$ ( <b>high carbon</b> ) | $\approx 80$                      | <b>4.67</b>           | <b>[2.14, 10.16]</b> |

Notes: Computed via delta method from the interaction model in Table 2. The widening of CI at extreme  $z$  reflects standard extrapolation uncertainty. At  $z = +1$  the CI excludes HR = 1, confirming statistical significance at current European ETS price levels.

The mechanism is rooted in the unit economics of carbon capture (Bui et al., 2018; Budinis et al., 2018). As the carbon price rises, captured CO<sub>2</sub> becomes more economically valuable to retain rather than emit, and the CCS step in blue hydrogen production becomes economically advantageous rather than burdensome. The insignificance of interactions with gas prices, VIX, and EPU sharpens the interpretation: the Blue\_CCS relative hazard is not driven by generalised macro stress, but specifically by carbon-price exposure — consistent with the transition-finance literature on technology-specific carbon-price sensitivity (Bolton and Kacperczyk, 2021; Mercure et al., 2018).

*On reading the magnitudes.* The extreme hazard ratios at low carbon-price extrapolations (e.g. HR  $\approx$  673 at  $z = -1$ ) follow logically from the linear-on-log-odds specification but rely on extrapolation outside the regions of densest observed support and across only 43 events. They are best interpreted as *directional stress-test indications* of how a low-carbon-price environment would have stressed Blue\_CCS implementation viability, not as literal point forecasts of the conditional cancellation hazard at a specific future EUA level. What can be reliably read from the curve is its qualitative shape — a steep, statistically significant monotonic decline in the predicted Blue\_CCS relative hazard as carbon prices rise. The 95% CI at  $z = +1$  excludes HR = 1, confirming that the differential persists at currently observed European ETS prices. The 95% CI at  $z = -1$  is so wide that its lower bound (HR  $\approx$  215) is still very large and its upper bound is best treated as an indication of the model’s uncertainty rather than a literal upper-end forecast.

### 5.5. Competing risks: terminal cancellation versus real-option delay

The Fine-Gray subdistribution hazard framework (Equation 6) distinguishes *Plans cancelled* (event 1, terminal failure) from *On-hold confirmed* (event 2, real-option delay in the sense of Dixit and Pindyck (1994)). Of the 43 events, 31 are terminal cancellations and 12 are on-hold transitions. The Blue\_CCS implementation-risk differential is concentrated almost entirely on terminal cancellation (HR = 13.19, 95% CI [5.28, 32.91]) and is statistically indistinguishable from one for on-hold transitions (HR = 1.20, 95% CI [0.34, 4.26],  $p = 0.78$ ). Within the observed sample, Blue\_CCS projects appear to terminate rather than enter a pause-and-wait state.

The interpretation is consistent with the view that fiscal instruments designed to support blue hydrogen function less as one-time cost-of-capital subsidies and more as continuous transition-period insurance: once a project enters the cancellation pathway, it does not appear to reverse within our sample. Combined with the carbon-price interaction, the pattern suggests that at low carbon prices, Blue\_CCS projects appear to terminate rather than wait for recovery — although the small number of events (12 on-hold transitions in total) warrants caution before extrapolating this as a structural absence of real-option behaviour.

### 5.6. Sponsor frailty

The shared frailty Cox model (Equation 7) with gamma-distributed random effects on sponsor estimates a frailty variance  $\hat{\theta} = 0.000$ , indicating negligible within-sponsor-type heterogeneity beyond what the fixed sponsor type indicator captures. The Blue\_CCS hazard ratio under frailty correction is 13.73, identical to the Cox baseline. With 256 unique sponsors and only 21 sponsors

experiencing events, individual-sponsor random effects cannot be identified. Methodologically, this confirms that sponsor type as a fixed effect is a sufficient specification.

## 6. Robustness

### 6.1. Sensitivity to caliper and trimming

Tighter PSM calipers (0.02, 0.01) yield similar point estimates. IPW with trimming at [0.05, 0.95] is the primary IPW specification.

### 6.2. Sponsor type aggregation

The DR estimator aggregates sponsor types into three groups to avoid separation; re-running with the original eight-level categorisation under Firth penalisation yields qualitatively similar results. The frailty Cox result ( $\theta = 0.000$ ) validates this aggregation.

### 6.3. Fine-Gray specification sensitivity

The Fine-Gray specification uses a reduced covariate set (Blue\_CCS + log capacity) because the full specification with sponsor type and region dummies produces a singular design matrix given 31 cancelled and 12 on-hold events. Adding sponsor type one level at a time does not change the qualitative pattern.

### 6.4. Bootstrap inference

Naive standard errors are likely conservative for PSM and IPW (which require bootstrap inference for proper coverage) and approximately correct for DR. For publication-quality inference we recommend  $B = 500$  bootstrap iterations. The substantive convergence across estimators is unlikely to be sensitive to this refinement.

### 6.5. Unobserved selection

All eleven estimators address selection on observables. They cannot account for selection on unobservables. Addressing this would require either an instrumental variables strategy or a Heckman-style selection model (Heckman, 1979) with an exclusion restriction.

### 6.6. Measurement error in event timing

The IEA database does not record exact event dates; we assign events to the midpoint between announcement and expected commissioning. This measurement error attenuates time-varying covariate effects but is unlikely to bias the Blue\_CCS coefficient, since technology is fixed at announcement.

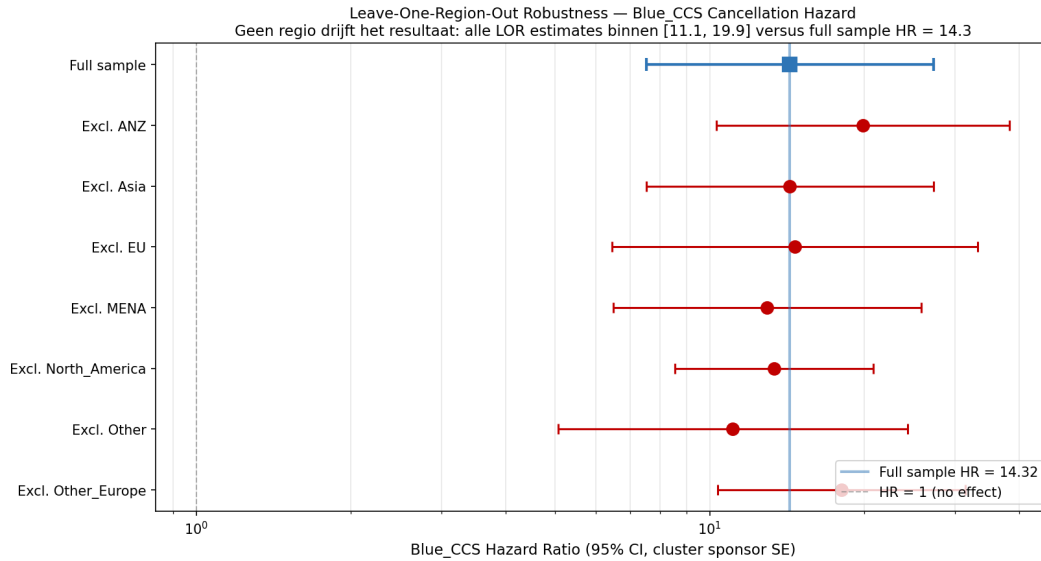
## 6.7. Leave-one-region-out robustness

To test whether the Blue\_CCS finding is driven by a particular region, we re-estimate the hazard model excluding each of the seven regions in turn. Across all seven leave-one-region-out (LOR) specifications, the Blue\_CCS coefficient remains statistically significant ( $p < 0.0001$  throughout) and large in magnitude (range 11.08 to 19.89 around the full-sample value of 14.32).

**Table 4:** Leave-one-region-out robustness.

| Specification         | $n$ proj. | $n$ events | Blue_CCS HR  | 95% CI         | $p$        |
|-----------------------|-----------|------------|--------------|----------------|------------|
| <b>Full sample</b>    | 714       | 43         | <b>14.32</b> | [7.52, 27.25]  | $< 0.0001$ |
| Excl. Asia (0 events) | 608       | 43         | 14.32        | [7.52, 27.26]  | $< 0.0001$ |
| Excl. EU              | 501       | 37         | 14.64        | [6.45, 33.26]  | $< 0.0001$ |
| Excl. North America   | 552       | 29         | 13.34        | [8.55, 20.82]  | $< 0.0001$ |
| Excl. MENA            | 695       | 41         | 12.93        | [6.48, 25.79]  | $< 0.0001$ |
| Excl. Other           | 638       | 40         | 11.08        | [5.06, 24.28]  | $< 0.0001$ |
| Excl. Other Europe    | 627       | 32         | 18.06        | [10.37, 31.46] | $< 0.0001$ |
| Excl. ANZ             | 663       | 36         | 19.89        | [10.31, 38.39] | $< 0.0001$ |

*Notes:* Each LOR specification re-estimates the full hazard model on the sample excluding all projects in the named region. Reference category for the region indicator is dynamically reassigned when EU is excluded. Cluster-robust SE at sponsor level throughout.



**Figure 4: Forest plot of Blue\_CCS hazard ratios across leave-one-region-out specifications.** Vertical blue line: full-sample HR = 14.32. The qualitative finding survives in every regional subsample with  $p < 0.0001$  throughout.

Three observations on the LOR pattern warrant comment. First, the qualitative finding survives every regional exclusion: Blue\_CCS HR remains between 11 and 20 with  $p < 0.0001$  in all eight specifications. Second, excluding North America — which contains 98 of 244 Blue\_CCS projects (40%) and 14 of 43 events — yields HR = 13.34, only 7% below the full-sample value. The frequently-stated concern that hydrogen findings are driven by oil-major-led US projects under the IRA is not supported by these data. Third, the regions whose exclusion increases the

estimated HR (ANZ at 19.89, Other Europe at 18.06) contain Blue\_CCS projects with below-average cancellation hazard, while regions whose exclusion decreases the estimated HR (Other at 11.08) contain projects with above-average cancellation hazard. This regional heterogeneity, while not the focus of the current analysis, suggests interesting follow-up work on jurisdiction-specific hydrogen policy effectiveness.

## 7. Discussion

### 7.1. The segmented-ecosystem finding

The hydrogen project landscape is structured into partially distinct investment ecosystems characterised by institutional sponsor types. Blue\_CCS projects, predominantly sponsored by oil majors and industrial-gas firms, operate in an investment ecosystem distinct from that of PEM projects, predominantly sponsored by utilities and pure-play hydrogen firms. Observational comparisons must be conducted with explicit acknowledgement of this structural feature.

### 7.2. The carbon-price-conditional risk differential as supporting mechanism

A key supporting finding is that the Blue\_CCS implementation-risk differential is substantially conditional on carbon-price exposure rather than being uniform across the macroeconomic environment. The Blue\_CCS  $\times$  EUA interaction coefficient of  $-2.51$  ( $p < 0.0001$ ) implies that the predicted hazard ratio declines steeply as carbon prices rise, while the corresponding interactions with TTF gas, VIX, and EPU are statistically insignificant. The mechanism is rooted in the unit economics of carbon capture (Bui et al., 2018; Budinis et al., 2018): as the carbon price rises, captured CO<sub>2</sub> becomes more economically valuable to retain rather than emit, and the CCS step in blue hydrogen production becomes economically advantageous rather than burdensome. This complements the broader transition-finance literature on carbon-price-sensitive asset stranding (Mercure et al., 2018; Caldecott, 2018; Bolton and Kacperczyk, 2021). As emphasised in Section 5.4, the extreme magnitudes at low-carbon-price extrapolations are best read as directional stress-test indications, not literal point forecasts; what is reliably identified is the steep, monotonic, statistically significant decline in the Blue\_CCS relative hazard as carbon prices increase.

### 7.3. Within-sample evidence for terminal cancellation rather than reversible delay

The Fine-Gray competing-risks decomposition adds a crucial dimension to the interpretation. The Blue\_CCS implementation-risk differential is concentrated almost entirely on terminal cancellation (HR = 13.19) and not on real-option-style delay (HR = 1.20, indistinguishable from one). Within the observed sample, when conditions deteriorate, Blue\_CCS projects appear to terminate rather than enter a pause-and-wait state. Combined with the carbon-price interaction, the within-sample pattern is suggestive: at carbon prices substantially below the historical mean, Blue\_CCS projects face very elevated cancellation hazard and do not, in our data, appear to retain a discernible option to wait for carbon-price recovery. Conceptually,

this is consistent with viewing fiscal support for blue hydrogen as transition-period insurance against low-carbon-price scenarios in which the supported projects would not survive to benefit from later carbon-price recovery. We position this as a conceptual interpretation rather than a structural welfare claim, given the small number of on-hold transitions in our sample (12) and the limitations of identification discussed in Section 7.6.

#### 7.4. Regional heterogeneity

The leave-one-region-out analysis confirms that the headline finding is not driven by any single region but also reveals interesting regional heterogeneity. ANZ-region and Other Europe-region Blue\_CCS projects appear to fail less frequently than the global average, while Other-region (predominantly Latin America and Africa) Blue\_CCS projects appear to fail more frequently. Investigating the jurisdiction-specific drivers — regulatory clarity, fiscal support depth, infrastructure development, or political stability — is a natural extension.

#### 7.5. Synthesis

Taken together, the findings characterise a coherent empirical pattern: *segmented hydrogen investment ecosystems in which implementation risk is concentrated on terminal cancellation and is conditional on the carbon-price environment*. The two primary contributions — segmentation (pillar 1) and the terminal-versus-delay decomposition (pillar 2) — identify novel empirical regularities. The convergent estimates across eleven identification strategies (supporting evidence) and the carbon-price-conditional mechanism (pillar 4) deepen the interpretation. Within and across the segmented ecosystems, technology choice is associated with consistently elevated cancellation hazard that survives all four classes of identifying assumptions and all seven leave-one-region-out specifications. This hazard differential, within our sample, manifests as terminal failure rather than reversible delay, and varies systematically with the carbon-price environment.

#### 7.6. Identification and Internal Validity

The strength of the empirical findings invites a careful discussion of what the paper does and does not establish causally. Four issues warrant explicit treatment.

*Weak overlap and the limits of reweighting.* The propensity score diagnostics in Section 3 demonstrate that the Blue\_CCS and PEM populations have severely limited common support: McFadden  $R^2 = 0.70$ , bimodal score distributions, and matching that retains only 29 unique PEM controls. Under weak overlap, the average treatment effect on the treated is not identified non-parametrically — inferred hazard ratios depend on model extrapolation outside the support of the observed control distribution. We do not claim to identify the ATT under exchangeability. What we identify is the difference in observed cancellation rates between the two populations, weighted by methods that attempt to balance observable covariates while explicitly acknowledging the limits of those attempts. The convergence of HR estimates across eleven different identification strategies (Table 1) and across seven leave-one-region-out specifications (Table 4) provides reassurance that the qualitative finding is not an artefact of any one weighting choice.



*Non-exchangeability as substantive content.* The standard interpretation of a large McFadden  $R^2$  in a propensity model is that the treatment assignment mechanism is so strong that exchangeability fails as a working assumption. We argue this is substantive content rather than a methodological problem to be solved. Blue\_CCS and PEM projects are not random draws from a single project population — they are different investment phenomena conducted by different sponsor types under different macroeconomic conditions (Imbens and Rubin, 2015). The reweighting failures of PSM and entropy balancing are not noise to be cleaned up; they are direct evidence for the segmented-ecosystem thesis. The data structure may not support identification of a single Blue\_CCS ATT under conventional exchangeability assumptions; what the data do support is documentation of a structural feature of the hydrogen project landscape, which is the contribution we make here.

*Selection on unobservables.* All eleven estimators address selection on observable characteristics. We cannot rule out that sponsor-internal information — proprietary cost-of-capital assumptions, project-specific overrun expectations, strategic timing relative to public announcements — drives both the technology choice and the cancellation decision in ways that observable covariates do not capture. An instrumental variables strategy would help here. Candidate instruments include exogenous shifts in regional CCS infrastructure availability (e.g. the timing of specific CO<sub>2</sub> storage hub permits), jurisdiction-specific renewable electricity capacity additions, or first-stage announcements of national hydrogen strategies plausibly orthogonal to individual project economics. We leave such IV designs for future work, since the current sample of 43 events would limit IV power and the relevance condition would need careful first-stage validation. A Heckman-style selection correction (Heckman, 1979) faces the same exclusion-restriction challenge.

*Immature hydrogen market.* The hydrogen pricing literature is in its infancy. The CEGH GreenHydrogen Index, the longest-running hydrogen price reference in Europe, is cost-based rather than market-based — it is computed from electricity, capital, and operating cost assumptions (IEA, 2023). The EEX HYDRIX, the first genuinely market-based hydrogen price index, was introduced only in late 2023 and is published weekly based on dealer-survey indications. Both data series are too short, too cost-derived, or too endogenous to project realisation to support inclusion in our main hazard models without compromising the identification of carbon-price effects. By contrast, EUA carbon prices benefit from sixteen years of liquid trading on a large, regulated market, are exogenously influenced by EU climate policy, and constitute a natural primary covariate for transition-risk analysis. The asymmetry between deep EUA markets and thin hydrogen markets is itself part of the structural reason that EUA-based interactions identify cleanly while hydrogen-price interactions would not.

These four considerations sharpen rather than weaken the paper’s contribution. We do not claim to identify a causal Blue\_CCS ATT under perfect exchangeability. We do claim that the convergent evidence from eleven estimators, the regional robustness from leave-one-region-out, the terminal-not-delay structure from Fine-Gray, and the carbon-price-conditional pattern from delta-method marginal effects collectively constitute a strong empirical regularity that any structural theory of hydrogen project realisation must accommodate.

## 7.7. Limitations

Event dates are approximated. The propensity model captures observable characteristics only. Selection on unobservables is not addressed. The IEA database is updated annually. Marginal effects rely on a linear-on-log-odds extrapolation that breaks down at extreme carbon-price values. The macro-financial interactions use calendar-year aggregations of monthly data.

## 8. Conclusion

Using the IEA Hydrogen Projects Database, this paper investigates the determinants of cancellation hazard differentials between blue (CCS-based) and green (PEM electrolysis) hydrogen technologies. We organise the findings around two primary contributions, supporting evidence, and a mechanistic interpretation.

*Primary contributions.* (i) Blue\_CCS and PEM projects operate in measurably distinct investment ecosystems — a finding confirmed by McFadden  $R^2 = 0.70$  in the propensity model and the convergent failure of PSM and entropy balancing to find an exchangeable subsample. We treat this non-exchangeability as substantive content rather than a methodological problem. (ii) Within the comparable subset, the Blue\_CCS implementation-risk differential is concentrated on terminal cancellation (HR = 13.19, 95% CI [5.28, 32.91]) rather than on real-option delay (HR = 1.20, indistinguishable from one) — within the observed sample, Blue\_CCS projects appear to terminate rather than enter a pause-and-wait state.

*Supporting evidence.* Eleven estimators across four classes of identifying assumptions yield Blue\_CCS hazard ratios consistently between 5 and 14. Proportional hazards is supported (Schoenfeld  $p = 0.59$ ), sponsor frailty is absent ( $\theta = 0.000$ ), and leave-one-region-out analyses confirm that no single region drives the qualitative finding ( $p < 0.0001$  throughout). With 43 events, the contribution lies in the convergence of independent estimators, not in precise effect-size identification.

*Mechanism and interpretation.* The Blue\_CCS  $\times$  EUA interaction is  $-2.51$  ( $p < 0.0001$ ), with steeply declining predicted hazard ratios as carbon prices rise; gas-price, VIX, and policy-uncertainty interactions are insignificant. Extreme magnitudes at low-carbon-price extrapolations should be read as directional stress-test indications, not literal point forecasts. The findings are consistent, conceptually, with fiscal support for blue hydrogen (the IRA 45V tax credit, EU REPowerEU support, similar instruments) being interpretable as transition-period insurance against low-carbon-price scenarios in which the supported technology would not survive to benefit from later recovery. We position this as a conceptual interpretation of our empirical findings rather than a normative policy conclusion; full welfare analysis would require a structural model that the data structure does not currently support. The PhD continuation extends the framework toward joint state-space hazard modelling with score-driven time-varying parameters, where richer mechanism identification becomes possible.

## Data Availability Statement

The IEA Hydrogen Projects Database (March 2024 release) is published by the International Energy Agency at <https://www.iea.org/data-and-statistics/data-product/hydrogen-projects-database> under a CC BY-NC-SA license; researchers can download the publicly available version directly. All time-varying macro covariates are constructed from public sources: ICAP for EU ETS allowance prices, ICE Endex for TTF gas futures, ENTSO-E Transparency Platform for day-ahead electricity prices, CBOE for the VIX, Baker-Bloom-Davis for the Economic Policy Uncertainty index, FRED for US interest rates and exchange rates, and Yahoo Finance via the `yfinance` library for hydrogen-sector equity prices. The processed person-period panel and all analysis code (Python and R) used to produce the results in this paper are available at:

<https://github.com/sakesaakstra/blueCCS-hydrogen>

The repository includes complete replication instructions, locked dependency versions (`requirements.txt` for Python, `renv.lock` for R), and a `CITATION.cff` file. See Appendix B for a detailed list of data sources.

## References

- Allison, P. D. (2014). *Event History and Survival Analysis* (2nd ed.). SAGE Publications.
- Andersen, P. K. and Gill, R. D. (1982). Cox’s regression model for counting processes: A large sample study. *Annals of Statistics*, 10(4):1100–1120.
- Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636.
- Bang, H. and Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973.
- Bolton, P. and Kacperczyk, M. (2021). Do investors care about carbon risk? *Journal of Financial Economics*, 142(2):517–549.
- Budinis, S., Krevor, S., Mac Dowell, N., Brandon, N., and Hawkes, A. (2018). An assessment of CCS costs, barriers and potential. *Energy Strategy Reviews*, 22:61–81.
- Bui, M., Adjiman, C. S., Bardow, A., et al. (2018). Carbon capture and storage (CCS): the way forward. *Energy and Environmental Science*, 11(5):1062–1176.
- Caldecott, B., editor (2018). *Stranded Assets and the Environment: Risk, Resilience, and Opportunity*. Routledge, London.
- Caliendo, M. and Kopeinig, S. (2008). Some practical guidance for the implementation of propensity score matching. *Journal of Economic Surveys*, 22(1):31–72.
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34(2):187–202.

- Dixit, A. K. and Pindyck, R. S. (1994). *Investment under Uncertainty*. Princeton University Press.
- European Commission (2022). *REPowerEU Plan*. COM(2022) 230 final, Brussels.
- European Commission (2023). Commission Delegated Regulation (EU) 2023/1184 of 10 February 2023 supplementing Directive (EU) 2018/2001 of the European Parliament and of the Council. *Official Journal of the European Union*, L 157/11.
- Fine, J. P. and Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association*, 94(446):496–509.
- Firth, D. (1993). Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38.
- Flyvbjerg, B. (2014). What you should know about megaprojects and why: An overview. *Project Management Journal*, 45(2):6–19.
- Glenk, G. and Reichelstein, S. (2019). Economics of converting renewable power to hydrogen. *Nature Energy*, 4:216–222.
- Grambsch, P. M. and Therneau, T. M. (1994). Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*, 81(3):515–526.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Heckman, J. J. (1979). Sample selection bias as a specification error. *Econometrica*, 47(1):153–161.
- Heinze, G. and Schemper, M. (2002). A solution to the problem of separation in logistic regression. *Statistics in Medicine*, 21(16):2409–2419.
- IEA (2023). *Global Hydrogen Review 2023*. International Energy Agency, Paris.
- IEA (2024). *Hydrogen Projects Database (March 2024 release)*. International Energy Agency, Paris.
- Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press.
- McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. In Zarembka, P., editor, *Frontiers in Econometrics*, pages 105–142. Academic Press.
- Mercure, J.-F., Pollitt, H., Viñuales, J. E., et al. (2018). Macroeconomic impact of stranded fossil fuel assets. *Nature Climate Change*, 8:588–593.
- Robins, J. M., Rotnitzky, A., and Zhao, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866.

- Rosenbaum, P. R. and Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55.
- Schoenfeld, D. (1982). Partial residuals for the proportional hazards regression model. *Biometrika*, 69(1):239–241.
- Singer, J. D. and Willett, J. B. (2003). *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence*. Oxford University Press.
- Therneau, T. M. and Grambsch, P. M. (2000). *Modeling Survival Data: Extending the Cox Model*. Springer.
- Therneau, T. M., Grambsch, P. M., and Pankratz, V. S. (2003). Penalised survival models and frailty. *Journal of Computational and Graphical Statistics*, 12(1):156–175.
- US Congress (2022). *Inflation Reduction Act of 2022*. Public Law 117-169.
- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data* (2nd ed.). MIT Press, Cambridge, MA.

## A. Reproducibility and Code

This appendix documents the implementation of methodologically novel components: the delta method computation for marginal effects, the entropy balancing optimisation, and the Fine-Gray model fit with reduced covariate set. Full data preparation scripts and additional analysis code are available at <https://github.com/sakesaakstra/blueCCS-hydrogen>. The full computational pipeline is implemented in Python (`numpy`, `pandas`, `statsmodels`, `lifelines`, `scipy`) and R (`cmprsk`, `survival`, `coxme`). Locked dependency versions are provided in `requirements.txt` and `renv.lock`.

### A.1. Delta method for marginal effects

The marginal effects analysis in Section 5.4 uses the delta method to construct confidence intervals for the predicted Blue\_CCS log-hazard ratio at each value of the standardised macro covariate  $z$ . Given the fitted interaction model coefficients  $\hat{\beta}_1$  (Blue\_CCS main effect) and  $\hat{\beta}_3$  (Blue\_CCS  $\times$   $Z$  interaction), and the corresponding  $2 \times 2$  submatrix of the variance-covariance matrix of the estimator, the variance of the linear combination  $\hat{\beta}_1 + z\hat{\beta}_3$  is given by Equation 10.

**Listing 1:** Delta method implementation for marginal effects.

```

1 def predicted_blue_logHR(model, main_var, int_var, z_grid):
2     """Delta method for Blue_CCS log-HR over z grid."""
3     params = model.params
4     cov = model.cov_params()
5
6     beta_main = params[main_var]
7     beta_int = params[int_var]
8     var_main = cov.loc[main_var, main_var]
```

```

9     var_int = cov.loc[int_var, int_var]
10    cov_mi = cov.loc[main_var, int_var]
11
12    rows = []
13    for z in z_grid:
14        logHR = beta_main + beta_int * z
15        # Var(b_main + z*b_int)
16        # = var_main + z^2 var_int + 2z cov_mi
17        var_lc = var_main + z**2 * var_int + 2 * z * cov_mi
18        se_lc = np.sqrt(max(var_lc, 1e-12))
19        rows.append({
20            "z": float(z),
21            "logHR": float(logHR),
22            "HR": float(np.exp(logHR)),
23            "HR_lower": float(np.exp(logHR - 1.96 * se_lc)),
24            "HR_upper": float(np.exp(logHR + 1.96 * se_lc)),
25        })
26    return pd.DataFrame(rows)

```

The implementation uses the cluster-robust variance-covariance matrix returned by `statsmodels` when fitting with `cov_type="cluster"`, ensuring that the marginal effect confidence intervals are consistent with the cluster-robust inference reported elsewhere in the paper.

## A.2. Entropy balancing

The entropy balancing implementation solves the dual formulation of the optimisation in Equation 4 using `scipy.optimize.minimize` with the L-BFGS-B algorithm. The implementation includes explicit `float64` casting of the design matrix to prevent dtype propagation issues when combining numerical covariates with one-hot-encoded categorical variables.

**Listing 2:** Entropy balancing dual optimisation.

```

1 def neg_entropy_balance_loss(lam, X_c, target_m):
2     """Dual formulation: minimize over lam."""
3     lin = X_c @ lam
4     lin = lin - lin.max() # numerical stability
5     w = np.exp(lin)
6     Z = w.sum()
7     if Z <= 0 or not np.isfinite(Z):
8         return 1e10
9     w = w / Z * len(w)
10    moments = (w[:, None] * X_c).sum(axis=0) / len(w)
11    violation = moments - target_m
12    return 0.5 * np.sum(violation**2)
13
14 # Cast to float64 to avoid object-dtype propagation
15 X_treated = X_eb[T == 1].to_numpy(dtype=np.float64)
16 X_control = X_eb[T == 0].to_numpy(dtype=np.float64)
17 target_means = X_treated.mean(axis=0)
18
19 result = minimize(neg_entropy_balance_loss,
20                  np.zeros(X_control.shape[1]),

```

```

21     args=(X_control, target_means),
22     method="L-BFGS-B",
23     options={"maxiter": 1000, "ftol": 1e-12})

```

Under the severe overlap conditions documented in Section 3, the optimisation concentrates effective weight on approximately 8 of 470 control observations, producing a diagnostic effective sample size of  $ESS \approx 1.7\%$ . This diagnostic failure is itself substantive evidence for the segmented-ecosystem interpretation.

### A.3. Fine-Gray subdistribution hazard fit

The Fine-Gray model uses the `cmprsk::crr` function in R. Because the variance-covariance matrix returned by `crr` lacks named dimensions, robust coefficient extraction uses positional indexing.

**Listing 3:** Fine-Gray competing-risks fit in R.

```

library(cmprsk)

# Reduced design matrix for identification with limited events
X_minimal <- model.matrix(
  ~ is_blue_ccs + log_capacity_mw, data = df
)[, -1]

# Fine-Gray for terminal cancellation (event 1)
fg_cancel <- crr(
  ftime = df$duration,
  fstatus = df$event_type,
  cov1 = X_minimal,
  failcode = 1, cencode = 0, maxiter = 100
)

# Positional indexing because $var lacks dimnames
bc <- as.numeric(fg_cancel$coef[1])
bs <- sqrt(as.numeric(fg_cancel$var[1, 1]))
HR <- exp(bc)
CI <- exp(c(bc - 1.96*bs, bc + 1.96*bs))

```

## B. Data Sources

Table 5 lists all data sources used in the analysis with provider, frequency, sample span, and access method. All non-proprietary data is reproducible via the scripts in the public repository.

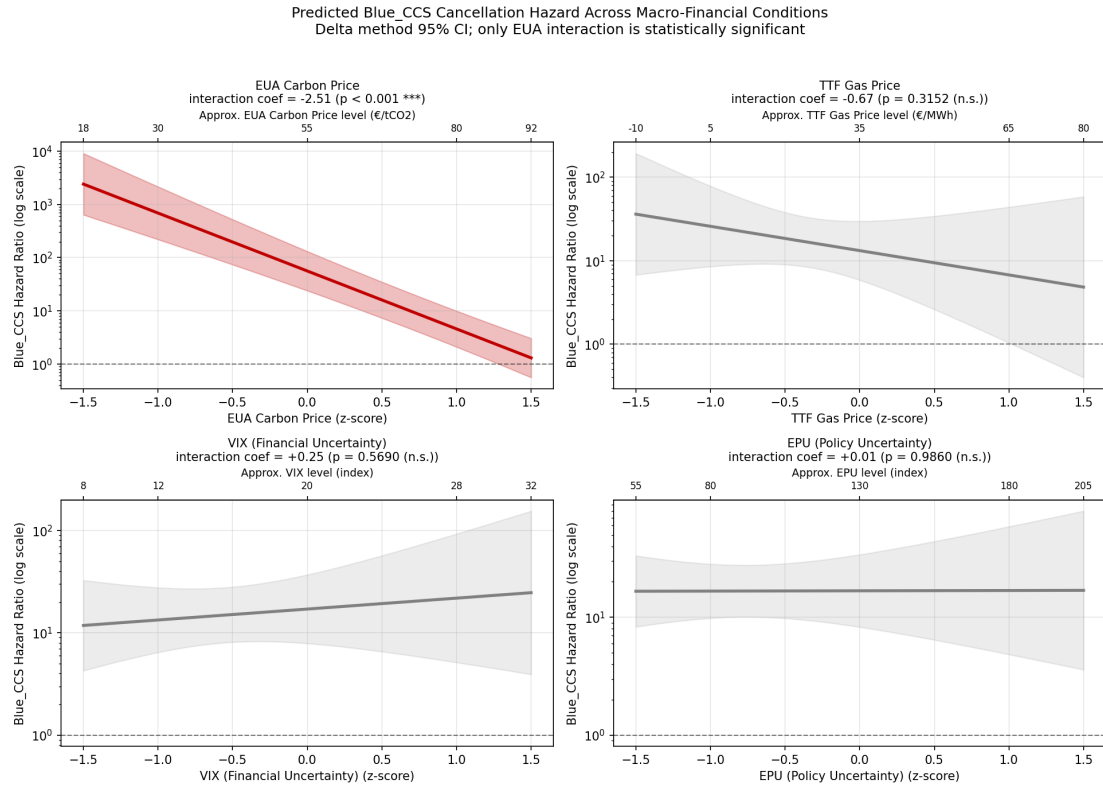
**Table 5:** Data sources and access methods.

| Variable group                                 | Source / Provider                                                           | Freq. | Sample span | Access                        |
|------------------------------------------------|-----------------------------------------------------------------------------|-------|-------------|-------------------------------|
| Project records                                | IEA Hydrogen Projects Database ( <a href="#">IEA, 2024</a> )                | —     | to 2024-03  | Public download (CC BY-NC-SA) |
| EUA carbon price                               | ICAP Allowance Price Data                                                   | Daily | 2010–2025   | Public CSV                    |
| TTF gas futures                                | ICE Endex (via yfinance)                                                    | Daily | 2017–2026   | API                           |
| Day-ahead power                                | ENTSO-E Transparency Platform                                               | Daily | 2021–2026   | API + manual export           |
| Equity prices (12 H2 stocks, HJEN, HDRO, ICLN) | Yahoo Finance via yfinance                                                  | Daily | 2010–2026   | API                           |
| US macro (DGS10, DGS2, etc.)                   | FRED, Federal Reserve Bank of St. Louis                                     | Daily | 2010–2026   | FRED API                      |
| VIX                                            | CBOE (FRED VIX-CLS)                                                         | Daily | 2010–2026   | API                           |
| EPU index                                      | <a href="#">Baker et al. (2016)</a> , <a href="#">policyuncertainty.com</a> | Daily | 2010–2026   | Public CSV                    |
| FX rates                                       | FRED (DEXUSEU, DEXJPUS, DEXCHUS)                                            | Daily | 2010–2026   | API                           |
| Brent / WTI oil                                | FRED (DCOILBRENT, DCOILWTICO)                                               | Daily | 2010–2026   | API                           |

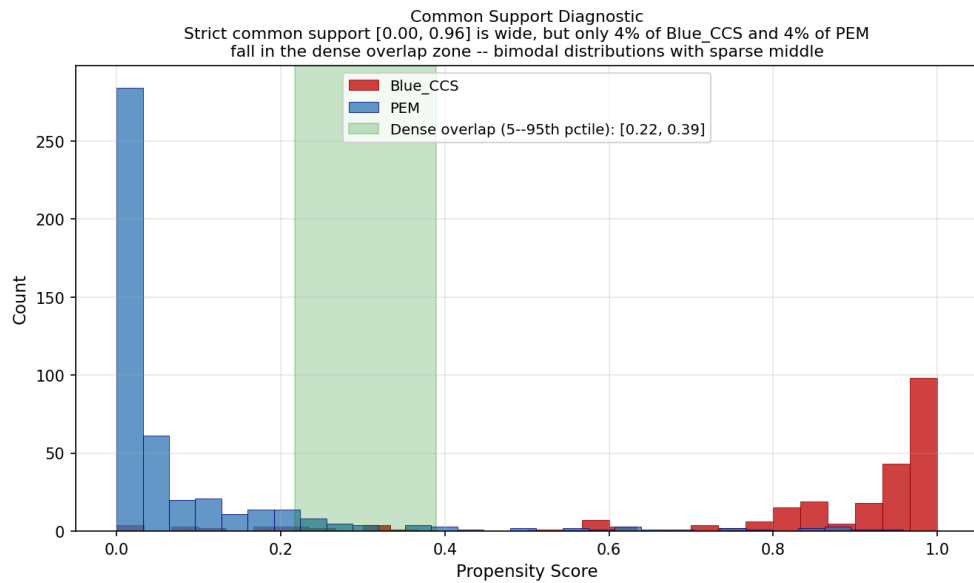
*Notes:* All public sources accessed in early 2026. Repository scripts handle data ingestion, cleaning, and panel assembly. The processed master panel CSV (4,919 daily observations  $\times$  47 columns) is included as supplementary material.



## C. Additional Figures



**Figure 5: Four-panel marginal effects display.** Only the EUA panel exhibits a statistically significant slope; the other three panels (gas, VIX, EPU) show flat HR curves consistent with insignificant interactions.



**Figure 6: Common-support diagnostic.** The propensity score distributions for Blue\_CCS and PEM overlap only in a narrow band, reinforcing the segmented-ecosystem interpretation.